Surveyor Pro's attachment document

Surveyor Pro was a government project for developing an information system that included expert systems, object-oriented databases, and wireless communications. The system would provide instant, graphical information to help government surveyors do their jobs. For example, after a surveyor touched a map on the screen of handheld device, the system would prompt the surveyor to enter the type of information needed for that area. This system would help in planning and implementing many projects, from layering fiber-optic cable to installing water lines.

Basis of estimates

- This project was preceded by a detailed study and proof of concept to show that it was possible to develop the hardware and software needed by surveyors and link the new devices to existing information systems. The proof of concept project produced a prototype handheld device and much of the software to provide basic functionality and link to the Global Positioning System (GPS) and other government databases used by surveyors. Some data is available to help estimate future labor costs, especially for the software development, and to help estimate the cost of the handheld devices.

- The main goal of this project is to produce 100 handheld devices, continue developing the software (especially the user interface), test the new system in the field, and train 100 surveyors in selected cities to use the new system. A follow-up contract is expected for a much larger number of devices based on the success of this project.
- The project has the following WBS:

1. Project management
2. Hardware
   - 2.1 Handheld devices
   - 2.2 Servers
3. Software
   - 3.1 Licensed software
   - 3.2 Software development
4. Testing
5. Training and support
6. Reserves

- Costs must be estimated by WBS and by month. The project manager will report progress on the project using earned value analysis, which requires this type of estimate.
- Costs will be provided in U.S. dollars. Because the project length is one year, inflation will not be included.
- The project will be managed by the government's project office. The project will require a part-time project manager and four team members. The team members will help manage various parts of the project and provide their expertise in the areas of software development, training, and support. Their total hours will be allocated as follows: 25 percent to project management, 25 percent to software development, 25 percent to training and support, and 25 percent to non-project work.
- The project involves purchasing the handheld devices from the same company that developed the prototype device. Based on producing 100 devices, the cost rate is estimated to be $600 per unit. The project will require four additional servers to run the software required for the devices and for managing the project.
- The project requires purchased software licenses for accessing the GPS and three other external systems. Software development includes developing a graphical user interface for the devices, an online help system, and a new module for tracking surveyor performance using the device.
- Testing costs should be low due to the success of the prototype project. An estimate based on multiplying the total hardware and software estimates by 10 percent should be sufficient.
- Training will include instructor-led classes in five different locations. The project team believes it will be best to outsource most of the training, including developing course materials, holding the sessions, and providing

help desk support for three months as the surveyors start using their devices in the field.

- Because several risks are related to this project, include 20 percent of the total estimate as reserves.
- You must develop a computer model for the estimate so that you can easily change several inputs, such as the number of labor hours for various activities or labor rates.

Fortunately, the project team can easily access cost estimates and actual information from similar projects. A great deal of information is available from the proof of concept project, and the team can also talk to contractors from the past project to help them develop the estimate. Computer models are also available, such as a software-estimating tool based on function points. **Function points** are a means of measuring software size based on what the software does for end users. Function points are comprised of inputs, outputs, inquiries, internal data, and external interface data. Allen Albrecht initially defined this metric in the 1970s, and today it is the international standard used to measure software size. It is the most commonly used software size metric, followed by lines of code.[15]

Because the estimate must be provided by WBS and by month, the team first reviews a draft of the project schedule. The team decides to begin by estimating the cost of each WBS item and then determine when the work will be performed, even though costs may be incurred at different times than when the work is performed. The team's budget expert has approved this approach for the estimate. The team has further assumptions and information for estimating the costs for each WBS category:

1. *Project management:* Estimate based on compensation for the part-time project manager and 25 percent of the four team members' time. The budget expert for this project suggested using a labor rate of $100/hour for the project manager and $75/hour for each team member, based on working an average of 160 hours per month, full time. Therefore, the total hours for the project manager under this category are 960 (160/2 * 12 = 960). Costs are also included for the four project team members who are each working 25 percent of their time: a total of 160 hours per month for all project personnel (160 * 12 = 1,920). An additional amount for all contracted labor is estimated by multiplying 10 percent of the total estimates for software development and testing costs (10% * ($594,000 + $69,000)).
2. *Hardware*
   - 2.1 *Handheld devices*: 100 devices estimated by contractor at $600 per unit.
   - 2.2 *Servers*: Four servers estimated at $4,000 each, based on recent server purchases.
3. *Software*
   - 3.1 *Licensed software:* License costs will be negotiated with each supplier. Because there is a strong probability of large future contracts and great publicity if the system works well, costs are expected to be lower than usual. A cost of $200/handheld device will be used.

3.2 *Software development:* This estimate will include two approaches: a labor estimate and a function point estimate. The higher estimate will be used. If the estimates differ by more than 20 percent, the project will require a third estimation approach. The supplier who developed the proof of concept project will provide the labor estimate input, and local technical experts will make the function point estimates.

4. *Testing*: Based on similar projects, testing will be estimated as 10 percent of the total hardware and software cost.
5. *Training and support*: Based on similar projects, training will be estimated on a per-trainee basis, plus travel costs. The cost per trainee (100 total) will be $500, and travel will cost $700/day/person for the instructors and project team members. The team estimates that the project will require a total of 12 travel days. Labor costs for the project team members will be added to this estimate because they will assist in training and providing support after the training. The labor hours estimate for team members is 1,920 hours total.
6. *Reserves*: As directed, reserves will be estimated at 20 percent of the total estimate.